How Much Can a Temperature Log Tell Us About Spoiling Milk? A Loss-Budget Benchmark for Cold-Chain Monitoring

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ABSTRACT

A cold-chain monitor logs temperature, but a log is not a verdict: bacterial growth also depends on strain, starting load and lag, none of which a thermometer sees. We ask how much of the error in predicting milk spoilage from a temperature record is caused by biology, by the sensor, by the sampling interval and by the structure of the growth model. We build an open Monte Carlo benchmark in which a Baranyi–Roberts model with a square-root temperature law stands in for real milk, and an edge-computable Gompertz shelf-life accumulator tries to predict it from noisy, sampled records. Methods are compared by AUC with a paired bootstrap on identical trips, against an oracle and against simple rules. Parameters come from published experiments; the model reproduces a published fluctuating-temperature experiment to within -4 %, and when fitted to 51 ComBase records of natural spoilage flora it shows that growth parameters from inoculated isolates overpredict by a factor of 1.62. Fitted to 46 *Listeria monocytogenes* records, the same law predicts 53 records held out by condition with accuracy factor 1.37. Across five parameter sets, including this hazard case study, unobservable biology costs 0.27–0.40 AUC in the baseline run (0.09–0.34 for *Listeria*); the kinetic score beats a plain degree-hour count by at most 0.04; a PT1000 over a DS18B20 gains at most 0.02; and sampling hourly instead of every minute is indistinguishable in most settings. A Monte Carlo probability is better calibrated than a single alarm. We conclude that, for this problem, effort belongs in measuring biology, not in better thermometers, and we release code, tests and data.

1. Introduction

Milk is perishable in a precise way. Bacteria that grow slowly at 4 °C grow quickly at 15 °C, and every hand-off between plant, truck, hub, shop and home is a chance for the temperature to drift. A large body of work in predictive microbiology turns a temperature history into an estimate of growth [1, 2], and recent surveys describe sensors, loggers and digital twins built on it [3–5]. The practical question for a cold-chain monitor is narrower: given only a temperature record, how well can an inexpensive calculation tell whether milk has gone past a limit, and what stops it doing better?

The stakes are partly economic and partly about health. Post-pasteurisation contamination with psychrotrophic bacteria causes roughly half to three-fifths of spoilage of fluid milk in the United States [6], so warm handling wastes food. Foodborne disease more broadly is estimated to cause about 600 million illnesses and 420 000 deaths a year, with 175 000 deaths a year in the WHO South-East Asia region [7]. Those deaths are dominated by other agents and are not specific to milk. We use the figures as context for why temperature control matters, and we model growth only (spoilage flora in the main study and *Listeria monocytogenes* in a case study), not disease.

Recent reviews note that few studies predict end of shelf life, that endpoint definitions are inconsistent, that few models are validated beyond their development data, and that prediction intervals or probabilities are preferable to single estimates [8].

Closest prior work. The nearest studies are uncertainty and sensitivity analyses of shelf-life simulations. For fluid milk, Lau et al. built a Monte Carlo model of spoilage from post-pasteurisation gram-negative contamination and found that the maximum growth rate and the storage temperature had the largest effect on the share of containers over 20 000 CFU/mL [9]. An earlier model of psychrotolerant spore-formers at a constant 6 °C likewise found the maximum growth rate most influential [10]; its extension to temperatures that change along the supply chain ranked home storage temperature, transport temperature and initial spore concentration highest in a global sensitivity analysis, and was released as a digital tool [11]. Giannakourou and Taoukis propagated parameter uncertainty and cold-chain temperature variability through nested Monte Carlo loops for a frozen product [12], and later added raw-material variability for vitamin C loss in juice, where temperature was again the most important factor [13]. In all of these, temperature enters as an uncertain input drawn from a distribution, so it naturally ranks near the top.

On the monitoring side, field data from Swedish chains showed large gaps between dynamically predicted and printed shelf life, with retail display the weakest stage [14]. Mean kinetic temperature has been used to estimate remaining shelf life, with an interval from the extremes of the record, for chicken and low-temperature-pasteurised milk [15]. A neural network that estimates product temperature from vehicle and door data cut temperature error sharply; with a simple Q_{10} model, errors of 1 and 2 °C correspond to 10 and 20 % shelf-life error [16]. Coupling predictive microbiology with heat-transfer simulation, short failures shortened beef shelf life by up to 10 % and intermittent ones by up to 60 % [17], and cyclic abuse of chicken breast shifted the flora towards *Pseudomonas* and accelerated spoilage [18]. Reviews describe how often cold chains break [19], sensor-network work addresses where probes should sit [20], and a recent framework for pasteurised milk combines a growth model, a recurrent network and a language-model reasoner, treating sensor noise and initial load as sources of uncertainty [21].

These studies ask how variable the outcome is, which input moves it most, how much a given abuse costs, or how to build a decision system. They do not ask the question a temperature monitor faces: starting from a *measured* record and judged

against a known ground truth, how much of the ability to rank trips by spoilage is lost to unobserved biology, to sensor error, to the sampling interval and to the choice of rule? To our knowledge no study has reported such a decomposition, and that is what this paper provides. The two views fit together. When the temperature is unknown it dominates the spread of outcomes, as the sensitivity analyses above find; once it is logged, what is left is mostly biology, which is what our loss budget measures. What is new is the size of each loss relative to the others, and in particular how little a better sensor or faster sampling recovers.

Contributions. (1) An open, tested benchmark that separates four sources of error in predicting spoilage from a temperature record: unobservable biology, sensor error, sampling interval and model structure. (2) A loss budget measured against an oracle, with paired comparisons against simple rules. (3) Parameters tied to published data and two validation checks on real records: a reproduction of a fluctuating-temperature experiment, and fits to ComBase records for spoilage flora and for *Listeria monocytogenes*. (4) Design rules and a calibration comparison between a Monte Carlo probability and a single alarm.

2. Background

Secondary model. The square-root (Ratkowsky) law relates the maximum specific growth rate μ (per hour, natural-log basis) to temperature below the optimum,

$$\sqrt{\mu(T)} = b\,(T - T_{\min}), \qquad T > T_{\min}, \tag{1}$$

and is widely used for psychrotrophic spoilage bacteria. A University of Tasmania thesis fitted it to 68 growth curves of a fast-growing *Pseudomonas* in broth and milk ($T_{\min} = -7.7$ °C, $b = 0.030$) [22]. A study of *Pseudomonas fluorescens* in UHT milk reported growth rates at 4, 15 and 29 °C that, fitted with Eq. (1), give $b = 0.027$ and $T_{\min} = -9.0$ °C [23]. The two sources agree to within 15 % on generation time at 4 °C (5.6 h versus 6.4 h).

Primary models. The Baranyi–Roberts model [24] describes the log population $y = \ln N$ and a physiological state q:

$$\frac{dy}{dt} = \mu(T)\,\frac{q}{1+q}\left(1 - e^{y - y_{\max}}\right), \qquad \frac{dq}{dt} = \mu(T)\,q, \qquad h_0 = \ln\!\left(1 + \tfrac{1}{q_0}\right), \tag{2}$$

so that, at constant temperature, lag $\lambda = h_0/\mu$ and the product $\lambda\mu = h_0$ is constant for cells with the same history. It handles changing temperature naturally, because μ is evaluated along the trace. The Gompertz model, by contrast, has a closed-form time to reach a limit, which makes it attractive for a microcontroller.

Related milk-specific work. The Tasmanian thesis [22] already built and industry-tested a milk *Pseudomonas* temperature model with prototype software. Lin et al. measured growth under fluctuating temperature and observed extra lag after a shift when it fell within the lag phase [23]; Zwietering et al. examined growth after shifts and found extra lag of about a quarter of the lag at the new temperature [25]. We do not claim the idea of a temperature-driven shelf-life estimate.

3. Models and benchmark

Fig. 1 summarises the benchmark: a simulated world supplies the truth, and a monitor that sees only a sensed, sampled record tries to recover it.

3.1. Edge estimator

Let $\mathrm{SL}(T)$ be the time for a population at constant T to reach a limit N_L according to the modified Gompertz curve,

$$\mathrm{SL}(T) = \frac{h_0}{\mu(T)} + \frac{A}{\mu_{10}(T)\,e}\left[1 - \ln\!\left(-\ln\frac{L}{A}\right)\right], \tag{3}$$

with $\mu_{10} = \mu/\ln 10$, $A = \log_{10}N_{\max} - \log_{10}N_0$ and $L = \log_{10}N_L - \log_{10}N_0$. A reading T_i taken over an interval Δt_i consumes a fraction $\Delta t_i/\mathrm{SL}(T_i)$ of the shelf life, and the cumulative shelf life used is

$$\mathrm{CSL} = \sum_i \frac{\Delta t_i}{\mathrm{SL}(T_i)}, \qquad \text{limit reached when } \mathrm{CSL} \geq 1. \tag{4}$$

This takes one table lookup and one addition per reading. The estimator is deliberately a different primary model from the reference, so the comparison is not circular.

3.2. Reference and trips

Each simulated trip has a product-temperature trace on a one-minute grid. Base temperature is drawn from 2.5–5 °C. Excursions arrive at a mean rate of one per 144 h of horizon, peak at 12, 14.3, 17.5 or 30.7 °C (weights 0.30, 0.25, 0.30, 0.15) and last a lognormal time with median 60 min, clipped to 10–480 min. Product temperature follows ambient with a first-order time constant: 5–20 min for a low-mass pouch, 1–4 h for an insulated crate. A reference population grows along the trace by Eq. (2), with $T_{\min} \sim \mathcal{N}(\bar{T}_{\min}, \sigma_T^2)$, a lognormal multiplier on b with coefficient of variation $cv \in \{0, 0.10, 0.25\}$, h_0 uniform over a measured range, and a random starting load.

A trip is *spoiled* if the reference population exceeds the limit within the label horizon. The horizon is the time the estimator takes to reach the limit at constant 4 °C with nominal parameters, rounded up to a whole hour, as a printed shelf life would be. It was fixed from label balance before any method comparison was examined. Prevalence is therefore constructed (about 0.5 for spoilage, 0.08–0.38 for the *Listeria* case) and absolute predictive values must not be read as field rates.

3.3. Sensors, sampling and scores

Sensors add a per-trip offset, noise and quantisation: DS18B20 (offset within ±0.5 °C, noise 0.05 °C, step 0.0625 °C), PT1000 with MAX31865 (±0.15 °C, 0.03, 0.03125) and an ideal sensor. Sampling intervals are 1, 15 and 60 min plus an adaptive rule. Five scores are computed from the measured record: CSL; degree-hours above 4 °C; time above 5 °C; time above 8 °C; and maximum temperature. An *oracle* CSL uses each trip's true parameters and noise-free temperature and sets the ceiling. A *structure-risk* run re-imposes a quarter of the lag state whenever product temperature departs from, or returns to, its two-hour running mean by more than 4 °C, which the estimator is not told about [25].

3.4. Statistics

The metric is the area under the ROC curve (AUC), which depends on ranking and not on a threshold. Intervals use the bootstrap; differences between methods use a paired bootstrap

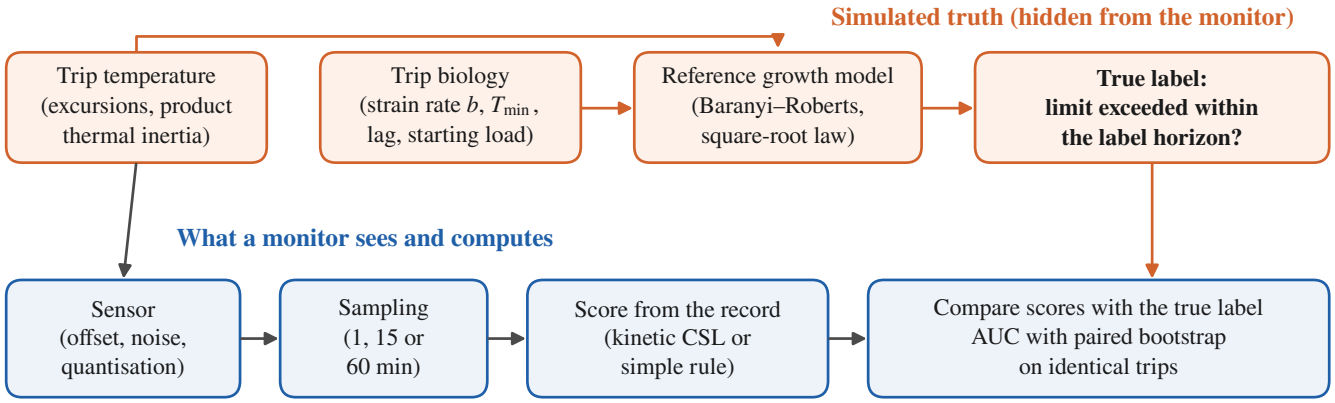


Fig. 1. Design of the benchmark. Each simulated trip has a true temperature history and a true biology, which the reference model turns into a true label (top). The monitor sees only a sensed, sampled record and turns it into a score (bottom). Scores are judged against the true labels by AUC; removing information step by step gives the loss budget.

Table 1
Benchmark runs. "Assumed" marks values without a source.

Run	Biology and labels
Baseline	$b = 0.030$, $T_{min} = -7.7$ °C [22]; $\sigma_T = 1.5$ K (assumed); $h_0 \in [2.5, 5]$ [23]; N_0 30–100 CFU/mL (assumed); limits 3×10^4, 10^6; horizons 83, 113 h
Extra lag	as baseline, plus extra lag after shifts [25]
Low load	as baseline, N_0 0.1–3 CFU/mL [6]; horizons 120, 154 h
Milk flora	$b = 0.0256$, $T_{min} = -6.6$ °C (ComBase, $n = 51$), low load; horizons 201, 258 h
Listeria	$b = 0.0235$, $T_{min} = -2.3$ °C (ComBase, $n = 46$); $h_0 \in [2, 4.5]$, $\sigma_T = 1.0$ K, N_0 0.06–0.18 CFU/mL (assumed); labels +0.5, +1 log; horizons 100, 202 h

on the same trips (4 000 trips per setting, 12 settings per run: two limits, two thermal regimes, three values of cv). An uncertainty-aware variant draws 80 parameter sets from a prior, computes CSL under each, and reports the fraction that reach the limit as a probability; it is scored by the Brier score and expected calibration error (ECE) against a single alarm with nominal parameters.

3.5. Parameter sets

Table 1 lists the five runs. The baseline uses the Neumeyer parameters and a measured lag range. Because inoculated-isolate parameters need not describe natural flora, a fourth run uses parameters fitted to ComBase records of spoilage flora (Section 4), and a fifth models *Listeria monocytogenes* with parameters fitted to ComBase records, with labels of +0.5 and +1 $\log_{10}$ growth above the starting load (round-number labels, not regulatory limits).

4. Validation against real data

Published experiment. Lin et al. held UHT milk inoculated with *P. fluorescens* at 4 °C with two four-hour excursions to 29 °C, and report that time to 10^6 CFU/mL fell from about 90 h to about 55 h [23]. With parameters fitted only to their growth rates and their 4 °C lag, our reference model gives 90 h and 53 h (-4 % on the excursion case, which was not used for fitting). The published times are approximate values read from the text; this is one experiment with one organism and an inoculated load far above real post-pasteurisation levels.

Natural spoilage flora. ComBase (Tasmanian Institute of Agriculture records) gives maximum growth rates of aerobic spoilage bacteria in pasteurised milk at 2.7–14.9 °C. The square-root law fits the 51 records well, with $b = 0.0256$, $T_{min} = -6.6$ °C and a residual standard deviation of 0.16 (natural-log units, including experimental noise). Parameters from inoculated isolates overpredict these records, with a bias factor of 1.62 that is nearly constant across temperature (Fig. 2): the temperature *shape* transfers, the *level* does not. This is consistent with studies that choose a fast isolate as a worst case.

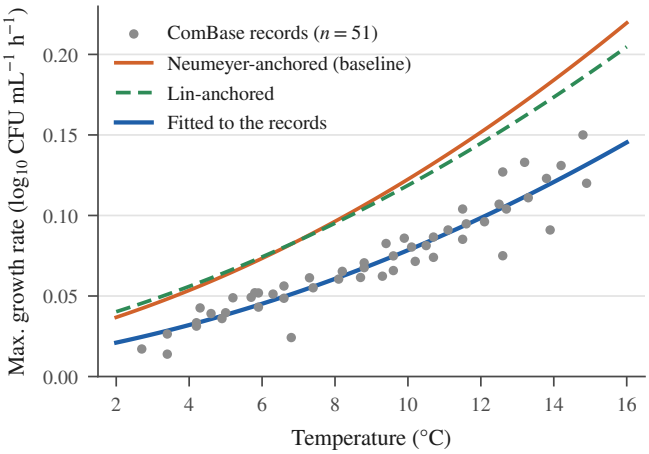


Fig. 2. Maximum growth rates of natural spoilage flora in pasteurised milk from ComBase (grey points) against the two literature-anchored models (orange, green) and the square-root fit to these records (blue).

***Listeria monocytogenes*.** Of the ComBase milk records, those with a stated rate for *L. monocytogenes* at 2–16 °C were split by condition: 46 records with no added factors were used for fitting and 53 records with other stated conditions (sterilised or previously heated milk, added salts or acids) were held out. The fit gives $b = 0.0235$ (95 % bootstrap interval 0.0203–0.0265) and $T_{min} = -2.3$ °C (-4.2 to -0.9), a generation time of 32 h at 4 °C, and a bias factor of 1.02 and accuracy factor of 1.37 on the held-out records (1.33 on the fitted ones). The held-out split is by condition, not by study, so some records share a source with the fitted ones and the test is weaker than a split by study. Because T_{min} is closer to refrigeration temperature, the rate at 13 °C is 5.9 times the rate at 4 °C for *Listeria*, against about 3.4 times for spoilage flora (Fig. 3).

Data provenance. The ComBase records come from the

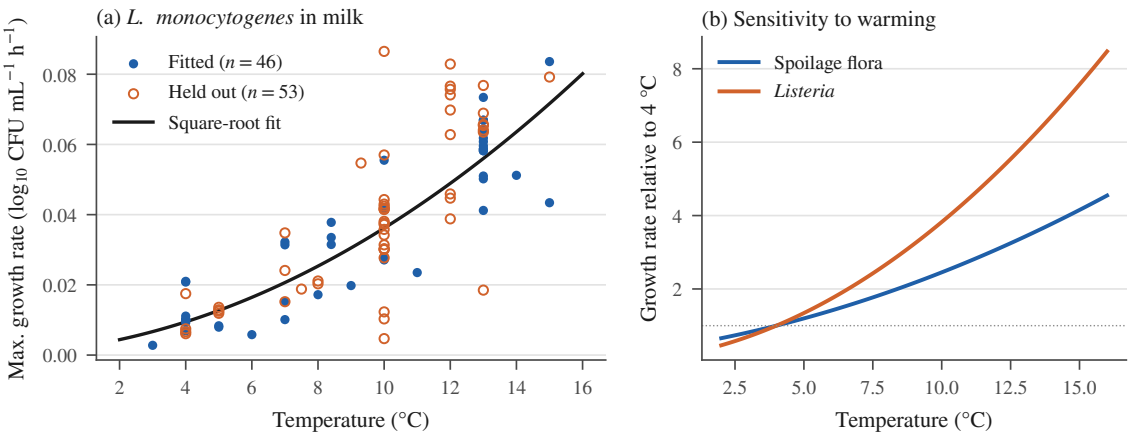


Fig. 3. *Listeria monocytogenes* in milk. (a) ComBase records used for fitting (filled), records held out by condition (open) and the square-root fit. (b) Growth rate relative to 4 °C for spoilage flora and *Listeria*: a warm hour costs the pathogen more.

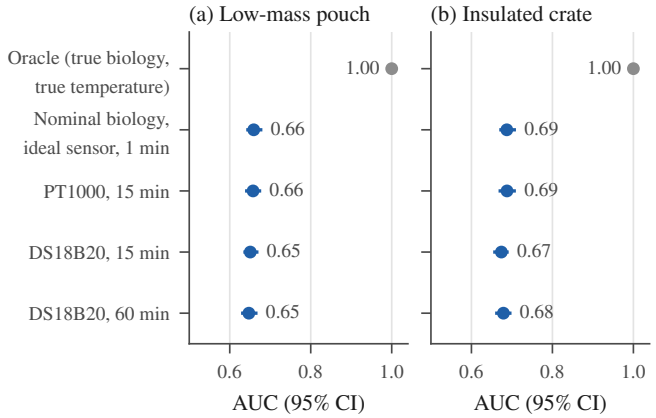


Fig. 4. Loss budget for (a) a low-mass pouch and (b) an insulated crate (limit 3×10^4 CFU/mL, $cv = 0.10$). Grey: oracle with true biology and true temperature. Blue: the kinetic score as information is removed step by step. Bars are 95 % bootstrap intervals.

database's search for milk. Of about 480 milk records, 373 were collected (179 with a stated growth rate), together with 51 records for aerobic spoilage flora in pasteurised milk. Records without a stated rate ("fit data" required) were not used.

5. Results

5.1. Loss budget

An oracle that knows each trip's true biology and temperature separates spoiled from unspoiled trips almost perfectly by construction. The same CSL with nominal biology and an ideal sensor loses 0.27–0.40 AUC across the 12 settings of the baseline run (Fig. 4). The loss is large even when strains do not differ in growth rate, because the minimum growth temperature, lag state and starting load still differ between trips. None of it is in the temperature record, so no sensor or sampling rate can recover it.

5.2. Against simple rules

On identical trips the kinetic score beats time above 5 °C in 12 of 12 settings and maximum temperature in 12. Against degree-hours above 4 °C, a rule that needs no growth model, the gain is positive with an interval excluding zero in 10 of 12 settings but small: 0.00–0.02 AUC in the baseline run and up to 0.04 when extra lag is present. Because both scores approximate the time integral of a growth rate, this is expected. Most of the value of the kinetic score is available from a degree-hour count.

5.3. Sensors and sampling

Replacing a DS18B20 with a PT1000 gains 0.00–0.02 AUC at 15-minute sampling. Sampling hourly instead of every minute shows no detectable difference in 11 of 12 settings; where it does, the effect is at most about 0.01. The adaptive rule is indistinguishable from fixed one-minute sampling. These results depend on the assumed excursion model (minimum duration 10 min, median 60 min) and would change if short, sharp excursions were common.

5.4. Robustness across parameter sets

Figure 5 summarises the four effects across all five runs. The pattern is the same whether the biology is anchored to inoculated isolates, includes extra lag, starts from a realistic low load, is fitted to natural flora, or describes *Listeria*: the biology gap is 0.28–0.40 (low load), 0.28–0.40 (milk flora) and 0.09–0.34 (*Listeria*); the gain over degree-hours stays below 0.04 in every run, and the sensor gain stays small (0.01–0.02 for *Listeria*).

5.5. Calibration

A single alarm using nominal parameters is badly calibrated against the constructed labels (ECE 0.34–0.43). A Monte Carlo probability with a prior matching the truth reaches ECE 0.03–0.07, and an overconfident prior on the growth rate 0.04–0.13. Absolute values reflect the constructed prevalence; the ranking of the three variants is the finding. The *Listeria* run needs a caveat. With the +1 log label the probability reaches ECE 0.08–0.20 against 0.34–0.41 for the alarm. With the half-log label the limit lies inside the early lag-to-growth transition, where the Gompertz estimator is a poor approximation of the reference; even the oracle reaches only 0.88–0.92 AUC there (1.00–1.00 for the +1 log label), and the probability, though still better than the alarm (ECE 0.28–0.41 against 0.42–0.45), is poorly calibrated in absolute terms. Label choice interacts with estimator structure.

6. Discussion

Where effort belongs. In every run, what the thermometer cannot see costs about an order of magnitude more accuracy than any choice of sensor or sampling interval. A better thermometer is therefore a poor way to improve a spoilage estimate; a cheap digital probe sampled every 15–60 min loses little under our assumptions. Effort is better spent on the biology: measuring starting load or a handling-history proxy,

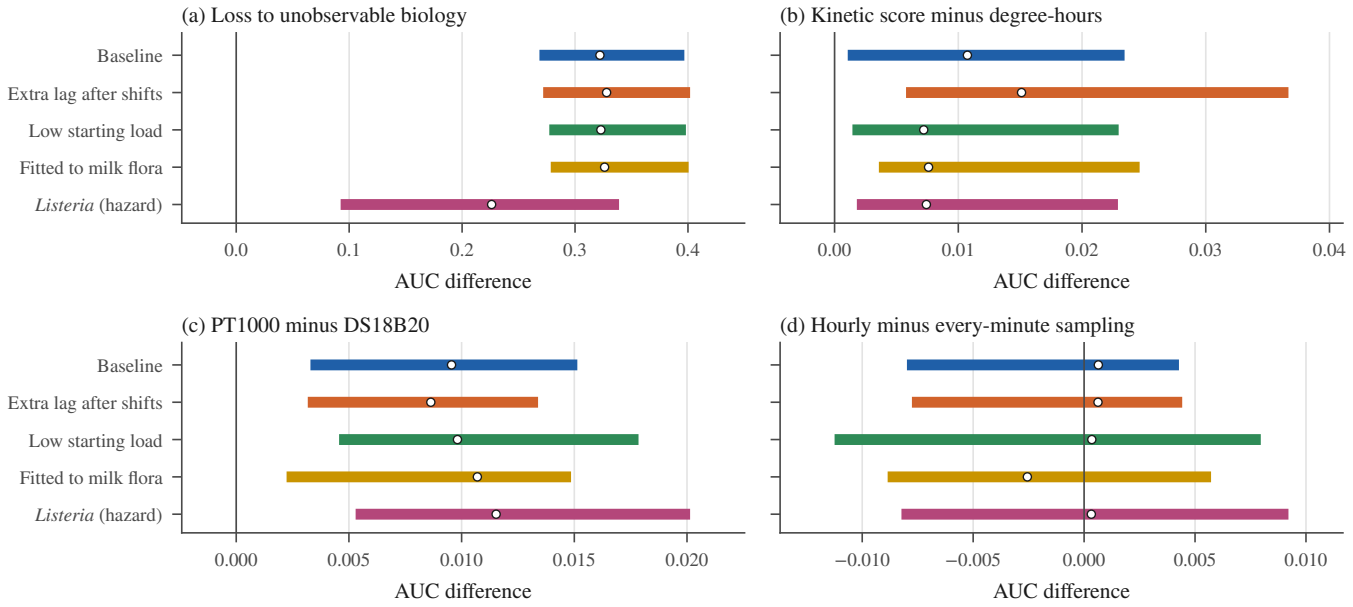


Fig. 5. Four effects across the five parameter sets, as AUC differences on identical trips: (a) loss to unobservable biology, (b) kinetic score over degree-hours, (c) PT1000 over DS18B20, (d) hourly over every-minute sampling. Bars span the 12 settings of each run; dots mark the median. Note the different horizontal scales.

learning the lag and strain spread for the product at hand, and reporting uncertainty instead of a single verdict.

Hazard versus spoilage. For spoilage flora, a warm hour is expensive but the baseline rate is high; for *Listeria*, growth at 4 °C is slow (generation time 32 h) but rises about 5.9-fold by 13 °C. The same benchmark and the same design rules hold for both, which supports treating temperature excursions as the common risk factor. This paper does not estimate illness risk, which depends on dose, virulence, host and the contamination event, none of which are modelled.

Limitations. (i) The reference and the estimator both turn the temperature history into something close to a time integral of growth, so the benchmark measures how much accuracy is lost to the sources listed, not whether a kinetic score models growth better; the structure-risk run relaxes one structural assumption only. (ii) Strain spread, starting load, excursion statistics and the *Listeria* lag are assumed. (iii) Labels are constructed, with prevalence near 0.5 (spoilage) and 0.08–0.38 (*Listeria*). (iv) ComBase records come from heterogeneous studies and methods; maximum rates are not curves, so lag and responses to shifts are untested on them. (v) One validation experiment reproduces fluctuating temperature, and no new plate counts were made. (vi) Neither organism class represents spore-formers, which account for a large share of spoilage [6]. (vii) For very small growth targets (the half-log *Listeria* label) the estimator's structure, not biology alone, limits accuracy.

7. Conclusion

A temperature log can say much about how much growth the milk could have had, and little about how much it did. In a tested, open benchmark anchored to published and ComBase data, unobservable biology dominates the loss; simple degree-hour counts come close to a growth model; sensor quality and sampling interval matter little; and a probability is better calibrated than an alarm. Direct measurement of biology and of fluctuating-temperature growth, with real plate counts, is the next step.

8. Reproducibility

All results are generated from fixed random seeds by scripts in the accompanying repository, with `make all` reproducing the tables and figures, 14 automated tests, and a parameter file with provenance for every number. Numbers in this paper are read from the result files by a script and not typed by hand.

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